

Word problems on the division of fractions

Deciding whether to multiply or divide — and finding the whole when a part is given.

LESSON 30–32

3 × 40 MIN

MATEMATIKA 6, §8

S7 7.5

LESSON CLOCK

25'

30'

30'

25'

10

Multiply or divide?	25 min
Finding the whole	30 min
A fraction of a fraction	30 min
Harder problems	25 min
Homework	10 min

LEARNING OBJECTIVES

- Decide from the wording whether a problem needs multiplication or division.
- Find the whole when a fractional part of it is known.
- Solve problems where a fraction of a fraction is taken.
- Check an answer against the sense of the problem.

TEXTBOOK

National	pp. 81–90
Cambridge	Stage 7, pp. 76–80
Workbook	Exercise 7.5

01

Explanation

Multiply or divide?

THE PROBLEM ASKS	OPERATION	EXAMPLE
a fraction of a known amount	multiply	$\frac{3}{4}$ of 20 = 15
how many small amounts fit in a large one	divide	$6 \div \frac{2}{3} = 9$
share an amount into equal parts	divide	$\frac{3}{4} \div 3 = \frac{1}{4}$
the whole, when a part is known	divide	$12 \div \frac{3}{4} = 16$

THE WORD “OF” MEANS MULTIPLY

And “how many fit”, “shared between” and “if this is a third of it” all mean divide. Reading for those phrases is more reliable than reading for the numbers.

Finding the whole

If $\frac{3}{4}$ of a length is 12 m, the whole is $12 \div \frac{3}{4} = 16$ m. Dividing by the fraction undoes taking the fraction.

GIVEN	WORKING	THE WHOLE
$\frac{3}{4}$ of it is 12	$12 \div \frac{3}{4}$	16
$\frac{2}{5}$ of it is 18	$18 \div \frac{2}{5}$	45
$\frac{5}{6}$ of it is 25	$25 \div \frac{5}{6}$	30
$\frac{1}{3}$ of it is 7	$7 \div \frac{1}{3}$	21

THE WHOLE IS ALWAYS BIGGER THAN THE PART

An answer smaller than the given part means the operations were the wrong way round. This single check catches nearly every error in this section.

A fraction of a fraction

PROBLEM	WORKING	ANSWER
$\frac{2}{3}$ of $\frac{3}{5}$ of a field	$\frac{2}{3} \cdot \frac{3}{5}$	$\frac{2}{5}$
half of $\frac{4}{7}$	$\frac{1}{2} \cdot \frac{4}{7}$	$\frac{2}{7}$
$\frac{3}{4}$ of a class of 28 are girls; $\frac{1}{3}$ of the girls wear glasses	$21 \cdot \frac{1}{3}$	7

"OF" CHAINS MULTIPLY

A fraction of a fraction of a whole is one product with three factors. There is no need to work out each stage separately unless the question asks for it.

Harder problems

PROBLEM	WORKING	ANSWER
a tank is $\frac{3}{5}$ full with 24 litres; its capacity	$24 \div \frac{3}{5}$	40
$\frac{2}{3}$ of a job is done in 4 days; the whole job	$4 \div \frac{2}{3}$	6 days
after spending $\frac{2}{5}$, 18 000 sum is left; the original	$18\,000 \div \frac{3}{5}$	30 000
$\frac{3}{8}$ of a road is 15 km; the rest	$40 - 15$	25 km

READ WHAT THE GIVEN NUMBER IS A FRACTION OF

In the third row 18 000 is what is **left**, so it is $\frac{3}{5}$ of the money, not $\frac{2}{5}$. Naming the fraction that matches the given number is the whole difficulty.

02

Worked examples

EXAMPLE 1 $\frac{3}{4}$ of a rope is 12 m. How long is the whole rope?

1

A part is given; the whole is wanted, so divide.

2

$12 \div \frac{3}{4} = 12 \cdot \frac{4}{3}$

3

$= 16$ m.

Bigger than 12 ✓

ANSWER

16 m

EXAMPLE 2 A tank is $\frac{3}{5}$ full and holds 24 litres. What is its capacity?

① 24 litres is $\frac{3}{5}$ of the capacity.

② $24 \div \frac{3}{5} = 24 \cdot \frac{5}{3}$

③ = 40 litres.

ANSWER

40 litres

EXAMPLE 3 After spending $\frac{2}{5}$ of her money, Nilufar has 18 000 sum left. How much had she?

① What is left is $1 - \frac{2}{5} = \frac{3}{5}$ of the money.

The key step.

② $18\ 000 \div \frac{3}{5}$

③ = 30 000 sum.

Check: $\frac{2}{5}$ of 30 000 is 12 000 ✓

ANSWER

30 000 sum

03 Interactive model

Give six problems and ask only for the operation, not the answer; deciding between $\times$ and $\div$ is the skill, and it is worth a lesson on its own.

Multiply or divide?

3 / 4 of 20:

multiply

divide

add

subtract

How many 2 / 3 m pieces fit in 6 m:

multiply

divide

add

subtract

3 / 4 of a rope is 12 m; the whole rope:

multiply

divide

add

subtract

The whole rope is:

9 m

16 m

15 m

48 m

A tank $\frac{3}{5}$ full holds 24 l; its capacity:

14.4 l

40 l

30 l

72 l

$\frac{2}{3}$ of $\frac{3}{5}$:

$\frac{2}{5}$

$\frac{5}{8}$

$\frac{10}{9}$

$\frac{6}{15}$ unsimplified

After spending $\frac{2}{5}$, what is left is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{5}$$

$$\frac{5}{3}$$

The whole must be:

smaller than the part

larger than the part

equal to it

either

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Word problem	Matnli masala	Текстовая задача
Part of a whole	Butunning qismi	Часть целого
To find the whole	Butunni topish	Найти целое
Fraction of a fraction	Kasrning kasri	Дробь от дроби
Remainder	Qoldiq	Остаток
Per unit	Bir birlikka	На единицу
To interpret	Talqin qilmoq	Истолковать
Sense check	Mantiqiy tekshirish	Проверка по смыслу

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Of” means:

“How many fit” means:

2 ⁵ of a number is 18; the number is:

5 6 of it is 25; it is:

30

20.8

35

150

2 3 of 3 5 equals:

$\frac{2}{5}$

$\frac{9}{10}$

$\frac{5}{8}$

$\frac{6}{8}$

A whole smaller than its part means:

a correct answer

the wrong operation

a large fraction

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{3}{4}$ of 20

ANSWER 15

2. $\frac{2}{3}$ of 27

ANSWER 18

3. 6 m in $\frac{2}{3}$ m pieces

ANSWER 9

4. $\frac{3}{4}$ of a rope is 12 m: the rope

ANSWER 16 m

5. $\frac{1}{3}$ of it is 7: it

ANSWER 21

6. $\frac{2}{5}$ of it is 18: it

ANSWER 45

7. $\frac{5}{6}$ of it is 25: it

ANSWER 30

Medium · the standard question

7 PROBLEMS

1. $\frac{2}{3}$ of $\frac{3}{5}$

ANSWER $\frac{2}{5}$

2. Half of $\frac{4}{7}$

ANSWER $\frac{2}{7}$

3. A tank $\frac{3}{5}$ full holds 24 l: its capacity

ANSWER 40 l

4. $\frac{2}{3}$ of a job in 4 days: the whole job

ANSWER 6 days

5. After spending $\frac{2}{5}$, 18 000 is left: the original

ANSWER 30 000

6. $\frac{3}{4}$ of 28 are girls; $\frac{1}{3}$ of them wear glasses

ANSWER 7

7. $\frac{3}{8}$ of a road is 15 km: the whole road

ANSWER 40 km

Hard · stretch and reason

7 PROBLEMS

1. $\frac{3}{8}$ of a road is 15 km: the rest

ANSWER 25 km

2. After giving away $\frac{1}{4}$ and then $\frac{1}{3}$ of the rest, 12 are left: the original

ANSWER 24

3. $\frac{4}{5}$ of a number is 4 more than $\frac{3}{5}$ of it: the number

ANSWER 20

4. A jug holds $1\frac{1}{2}$ l; how many $\frac{1}{4}$ l cups fill it?

ANSWER 6

5. $\frac{2}{3}$ of $\frac{3}{4}$ of 40

ANSWER 20

6. Half of a number added to a third of it is 25: the number

ANSWER 30

7. Why must the whole be larger than the part?

ANSWER The part is a fraction of it, under one

07 Homework

Homework — 5 tasks

Write the operation you have chosen before doing any arithmetic.

- $\frac{2}{3}$ of a length is 14 m. Find the whole length.
- A barrel is $\frac{5}{8}$ full with 30 litres. Find its capacity.

3. Find $\frac{3}{5}$ of $\frac{5}{9}$ of 36.
4. After spending $\frac{3}{7}$ of his money, Bekzod has 24 000 sum left. How much had he?
5. How many $\frac{3}{4}$ m planks can be cut from a board 9 m long?